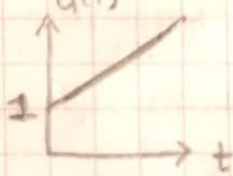


Finans Mat. Formüller

- Birikim fonksiyonu: $a(t)$ $a(0) = 1$
Sürekli Artan
- Tutar fonksiyonu: $A(t) = a(t) \cdot k$
- $(n-1, n]$ zaman aralığı = n . dönem
- $I_n = A(n) - A(n-1) = n$. dönemde kazanılan faiz
- Efektif faiz $= i_n = \frac{A(n) - A(n-1)}{A(n-1)} = \frac{I_n}{A(n-1)}$

- Basit faiz ve doğrusal birikim fonksiyonu:

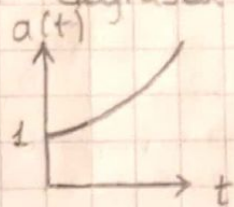
$$a(t) = 1 + it$$



$$i_n = \frac{a(n) - a(n-1)}{a(n-1)} = \frac{[1 + in] - [1 + i(n-1)]}{[1 + i(n-1)]} = \frac{i}{1 + i(n-1)}$$

- Bileşik faiz ve doğrusal olmayan birikim fonksiyonu

$$a(t) = (1+i)^t$$



$$i_n = \frac{a(n) - a(n-1)}{a(n-1)} = \frac{(1+i)^n - (1+i)^{n-1}}{(1+i)^{n-1}} = \frac{(1+i)^{n-1} \cdot i}{(1+i)^{n-1}} = i$$

- Nominal Faiz:

✓ Bir yıllık dönem sonunda efektif ve nominal faiz oranları üzerinden elde edilecek birikim değeri eşit.

$$1+i = \left(1 + \frac{i^{(m)}}{m}\right)^m$$

→ Buradan:

$$i(m) = m \cdot \left[(1+i)^{\frac{1}{m}} - 1 \right] \text{ çıkar.}$$

• İskontanın ölçümü:

✓ İskonto: ödünç verilen başlangıç tutarından, dönem içinde gerçekleşecek faiz geliri için peşin olarak ödünç verilen tutardan düşüldükten sonra geri kalının borca olarak verilmesi işleridir.

✓ Dönem sonunda 1 TL almak için dönem başında

$(1+i)^{-1}$ kadar yatırım yapılmalı.

✓ $v = (1+i)^{-1} = \text{İskonto Faktörü} = a^{-1}(t)$

✓ $a^{-1}(t) = \frac{1}{1+it}$ Basit faiz için iskonto fonksiyonu

$a^{-1}(t) = \frac{1}{(1+i)^t} = v^t \rightarrow \text{Bileşik faiz oranı için iskonto oranı}$

★ ✓ $(1+i)^t = \text{Birikim fon.} = 1 \text{ brlik yatırımın } t \text{ dönem sonunda birikimli değerini verir.}$

$v^t = 1 \text{ dönem sonunda ödenecek 1 birimin bugünkü değerini verir.} \rightarrow \text{Ya da peşin değer.}$

• Efektif İskonto:

$$d_n = \frac{A(n) - A(n-1)}{A(n)} = \frac{I_n}{A_n} \rightarrow n \text{ dönem için iskonto tutarı}$$

ef faiz

$$i = \frac{d}{1-d}$$

$$d = \frac{i}{1+i}$$

→ Bir kişi d iskonto oranı ile borca aldığında başlangıç değeri $1-d$ ve faiz tutarı da d dir.

Equivalency =)

effective → para ne kadar değişti
başlangı

Sh
disco

iskonto oranı → v → iskonto faktörü

$$d = i \cdot \frac{1}{1+i} = i \cdot v$$

$$d = \frac{i}{1+i} = 1 - \frac{1}{1+i} = 1 - v$$

iskonto oranı ve
iskonto faktörü ara-
sındaki ilişkiler.

Başla bir ilişki

$$d = i \cdot v = i \cdot (1 - d) = i - i \cdot d$$

$$i - d = i \cdot d$$

Basit iskonto

$$a^{-1}(t) = 1 - dt$$

→ Sabit basit faiz oranı, yatırım dönemi arttıkça efektif faiz oranı azalır. Basit iskonto oranı için tersi durum söz konusudur.

→ Basit ve bileşik iskonto oranları bir dönem için aynı sonuçları verirler. Daha uzun dönemler için basit iskonto, bileşik iskontoya göre daha az bugünkü değer oluşturma-
sına neden olur.

Bileşik iskonto:

$$a^{-1}(t) = (1 - d)^t$$

Nominal iskonto:

→ Bir dönem sonunda birikimli değeri 1 birim olan yatırımın bugünkü değerinin efektif iskonto ve nominal iskonto

Oran üzerinden bugünkü değeri eşittir

$$1+d = \left(1 + \frac{d^{(m)}}{m}\right)^m$$

$$d = 1 - \left(1 - \frac{d^{(m)}}{m}\right)^m$$

$$d^{(m)} = m \cdot \left[1 - \left(1 - d\right)^{1/m}\right] = m \cdot \left[1 - v^{1/m}\right]$$

→ 1 br'in nominal faiz oranı üzerinden bir dönem sonundaki birikimli değerinin, nominal iskonto oranı üzerinden birikimli değerine eşitliği,

$$\underbrace{\left[1 + \frac{i^{(m)}}{m}\right]^m}_{1+i} = \underbrace{\left[1 - \frac{d^{(p)}}{p}\right]^{-p}}_{1+i}$$

• Annuel Faiz ve Iskonto Oranı:

Çok küçük zaman dilimlerine ilişkin faiz ve iskonto oranları da incelenebilir. Bu oranlar annuel faiz oranları olarak adlandırılır.

$$\star S_t = \frac{A'(t)}{A(t)} = \frac{a'(t)}{a(t)} = \frac{d}{dt} \ln[A(t)] = \frac{d}{dt} \ln[a(t)]$$

Annuel faiz oranına göre 1 br'in t süre sonraki birikimli değerini bulunması için eşitlikte t yerine r yerleştirilerek her iki yarı [0, t] aralığında integrali alındığında,

each

δ = (constant) force of interest

δ_t = (general) force of interest t - measured in years.

$$\star \int_0^t \delta_r dr = \int_0^t \frac{d}{dr} \ln[a(r)] dr = \ln[a(t)] - \ln[a(0)]$$

$$= \ln \frac{a(t)}{a(0)}$$

$$\star \exp \left[\int_0^t \delta_r dr \right] = \frac{a(t)}{a(0)} = a(t)$$

- Birikim fonksiyonunun sabit basit faiz oranı ile hesaplanması durumunda anlık faiz oranı

$$\delta_t = \frac{\frac{d}{dt} a(t)}{a(t)} = \frac{\frac{d}{dt} (1+it)}{1+it} = \frac{i}{1+it} \quad 0 \leq t$$

- Birikim fonksiyonunun Sabit basit iskonto oranı kullanıldığında ise

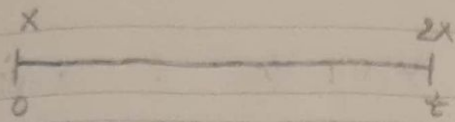
$$\delta_t = - \frac{\frac{d}{dt} a^{-1}(t)}{a^{-1}(t)} = - \frac{\frac{d}{dt} (1-dt)}{(1-dt)} = \frac{d}{1-dt} \quad 0 \leq t < \frac{1}{d}$$

- Anlık faiz ve iskonto oranı, nominal faiz oranı ve iskonto oranlarının tanımları kullanılarak bir yıl için faiz dönüşüm dönemi sayısının sonsuza gitmesi durumunda elde edilir:

$$\lim_{m \rightarrow \infty} i^{(m)} = \lim_{m \rightarrow \infty} d^{(m)} = \delta$$

- Suppose $a(t) = b^t$

$$\text{Then } \delta_t = \frac{a'(t)}{a(t)} = \frac{[b^t]'}{b^t} = \frac{b^t \ln(b)}{b^t} = \ln(b)$$



$$x \cdot (1.06)^t = 2x$$

$$t \ln(1.06) = \ln 2 \rightarrow t = \underline{\underline{11.89 \text{ years}}}$$

- If logarithms are not available, then use an interest table and perform a linear interpolation

↳ Johannine yöntemi

Rule 72 for doubling a single payment

$$n = \frac{\ln[2]}{\ln[1+i]}$$

$$n \approx \frac{0.72}{i}$$

Rule 114 for tripling a single payment

$$n = \frac{\ln[3]}{\ln[1+i]}$$

$$n \approx \frac{1.14}{i}$$

- An Approximate Approach for Multiple Payments

$$\left(\sum_{k=1}^n S_k \right) \cdot v^t = \sum_{k=1}^n S_k v^{t_k} \rightarrow v^t = \frac{\sum_{k=1}^n S_k v^{t_k}}{\sum_{k=1}^n S_k}$$

$$t \cdot \ln[v] = \ln x$$

$$t = \frac{\ln[A] - \ln[B]}{\ln[v]}$$

$$\bar{t} = \frac{S_1 \cdot t_1 + S_2 \cdot t_2 + \dots + S_n \cdot t_n}{S_1 + S_2 + \dots + S_n}$$

This approx. of t is always greater than value of t which means that

the present value using the met. of equated time < True present value

- ★ Simple interest is computed using the exact number of days for the period of investment and 365 as the number of days in a year. This is called exact Simple interest, and is denoted by "actual/actual".
- ★ The second method assumes that each month has 30 days and that the entire year has 360 days. Simple int. computed on this method is called ordinary simple interest and is denoted by "30/360".
- ★ The third method is a hybrid. It uses the exact number of days for the period of investment, but uses 360 days per year. Simple interest on this basis is called the banker's rule and is denoted by "actual/360".
- ★ The Banker's Rule is always more favorable to a lender than exact Simple int. and is usually more favorable than ordinary simple interest, but there are exceptions to that.
- ★ It is assumed, unless stated otherwise, that in counting days, interest is not credited for both the date of deposit and the date of withdrawal, but for only one of these dates.
- ★ Exact Simp. int $\rightarrow$ 365 days

Ordinary simp. int $\rightarrow$ One month equals to 30 days.

Banker's Rule $\rightarrow$ Count actual days where one year 360.

★ Whereas the choice of a comparison date has no effect on the answer obtained with compound interest, the same cannot be said of simple interest.

$\hookrightarrow$ Basit faizde karşılaştırma günü farkı, sonucu doğru olur.

Interpolasyon: Tahminleme

Bilinmeyen faiz oranının bulunması:

$\swarrow$
Doğrudan çözüm

$\downarrow$
Matematiksel tekniklerle

$\downarrow$
Faiz oran tabloları ile interpolasyon

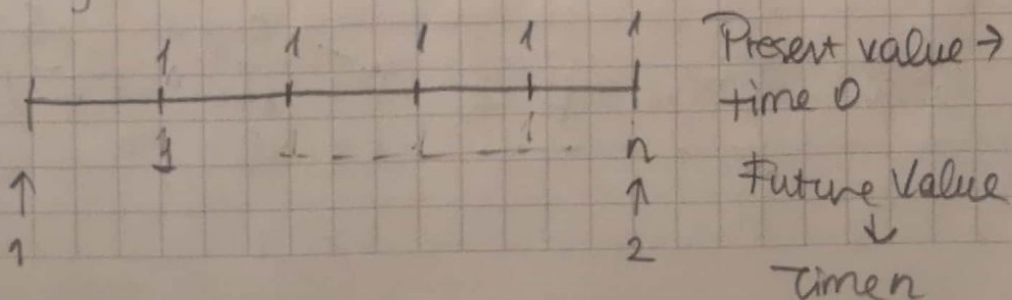
★ Değişken faiz oranı:

$$a(t) = (1+i_1) \cdot (1+i_2) \cdot (1+i_3) \cdots = \prod_{k=1}^t (1+i_k)$$

Annuity Immediate Dönem sonu annüite

Annuity: Series of payments made at equal intervals of time.

Annuity certain $\neq$ Contingent Annuity



of payments is n .

$a_{\overline{n}|}$ → Present value

$s_{\overline{n}|}$ → Accumulated Future value

$$\rightarrow a_{\overline{n}|} = 1.v + 1.v^2 + 1.v^3 + \dots + 1.v^n$$

$$v.(1+v+\dots+v^{n-1}) = v \cdot \frac{1-v^n}{1-v} = \frac{1}{1+i} \cdot \frac{1-v^n}{d}$$

$$= \frac{1}{1+i} \cdot \frac{1-v^n}{\frac{i}{1+i}}$$

$$P.V = a_{\overline{n}|} = \frac{1-v^n}{i}$$

$$\frac{i}{1+i} \rightarrow d$$

$$d = 1.v$$

$$\rightarrow s_{\overline{n}|} = 1 + (1)(1+i) + (1)(1+i)^2 + \dots + (1)(1+i)^{n-1}$$

$$\frac{1 - (1+i)^n}{1 - (1+i)} = \frac{(1+i)^n - 1}{i} = s_{\overline{n}|} = A.v$$

$$a + ar + ar^2 + \dots + ar^{n-1}$$

$$= \sum_{k=0}^{n-1} a.r^k = a \frac{1-r^n}{1-r}$$

$$s_{\overline{n}|} = a_{\overline{n}|} (1+i)^n$$

$$\frac{1}{a_{\overline{n}|}} = \frac{1}{s_{\overline{n}|}} + i$$

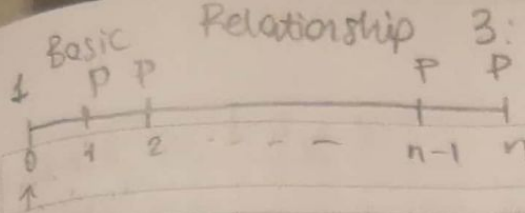
Basic Relationship 1: $1 = 1.v + v^n$

$$1.v + (1.v)^n$$

$$a_{\overline{n}|} = \frac{1-v^n}{i}$$

$$1 = i.a_{\overline{n}|} + v^n$$

Basic Relationship 2: $P.V. \cdot (1+i)^n = F.V.$ and $F.V. \cdot v^n = P.V.$



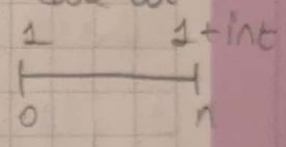
$$P \cdot a_{\overline{n}|i} = 1$$

$$P = \frac{1}{a_{\overline{n}|i}} \rightarrow \text{Sden msk}$$

$$\frac{1}{a_{\overline{n}|i}} = \frac{1}{s_{\overline{n}|i}} + i$$

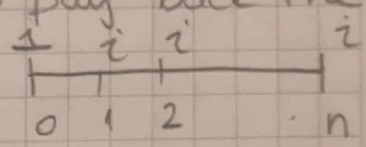
Interest Repayment Options

- Pay back the loan and all interest due at time n .



Total int Paid = $A(n) - A(0)$
 $\text{Loan} \cdot (1+i)^n - \text{Loan}$
 $\text{Loan} \cdot ((1+i)^n - 1)$

- Pay the end of each year, the interest that comes due on the loan and then pay back the loan at time n .

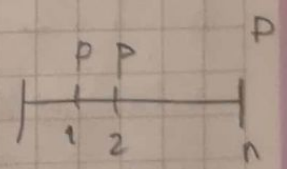


Annual interest pay: $i \cdot \text{Loan}$

Total int paid: $i \cdot \text{Loan} \times n$

- Pay a level annual amount at the end of each year for the next n years.

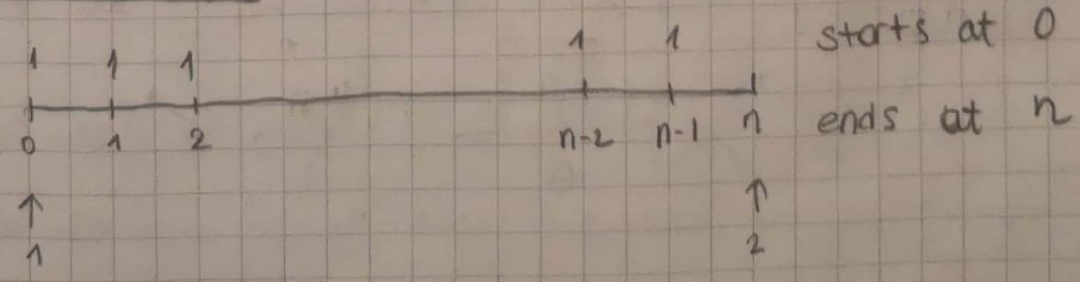
Annual payment $\frac{\text{Loan}}{a_{\overline{n}|i}}$



→ dene korti

Total payments $\left(\frac{\text{Loan}}{a_{\overline{n}|i}} \right) \times n$

Annuity Due



starts at 0
ends at n

$$a_{\overline{n}|} = 1 + 1 \cdot v + 1 \cdot v^2 + \dots + 1 \cdot v^{n-1} \rightarrow \text{Present Value}$$

$$a_{\overline{n}|} = \frac{1-v^n}{1-v} = \frac{1-v^n}{d} \quad a_{\overline{n}|} = \frac{1-v^n}{i}$$

$s_{\overline{n}|}$ → Accumulated value

$$= 1 \cdot (1+i) + 1 \cdot (1+i)^2 + \dots + (1+i)^n \cdot i$$

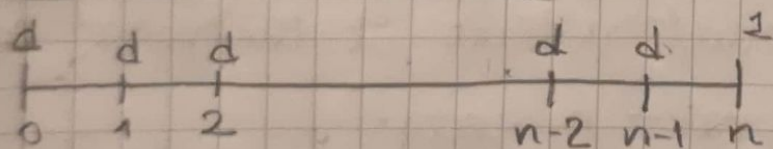
$$= 1+i \cdot (1 + (1+i) + \dots + (1+i)^{n-1})$$

$$1+i \cdot \left[\frac{1 - (1+i)^n}{1 - (1+i)} \right] = 1+i \cdot \left(\frac{(1+i)^n - 1}{i} \right)$$

$$s_{\overline{n}|} = \frac{(1+i)^n - 1}{d}$$

$$s_{\overline{n}|} = \frac{(1+i)^n - 1}{i}$$

• Basic Relationship 1:



$$1 = d \cdot a_{\overline{n}|} + v^n$$

The present value of this multiple payment income

stream at $t=0$ is $d \cdot a_{\overline{n}|} + (1) v^n = 1$

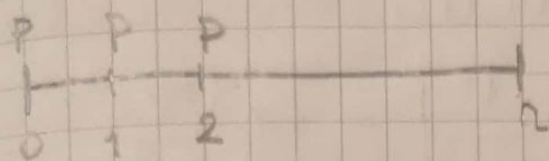
Note that $a_{\overline{n}|} = \frac{1-v^n}{d} \rightarrow 1 = d \cdot a_{\overline{n}|} + v^n$

• Basic Relationship 2: $PV(1+i)^n = FV$ and $PV = FV \cdot v^n$

$$s_{\overline{n}|} \cdot v^n = \left(\frac{(1+i)^n - 1}{d} \right) \cdot v^n \rightarrow a_{\overline{n}|} \cdot (1+i)^n = s_{\overline{n}|}$$

$$s_{\overline{n}|} \cdot v^n = a_{\overline{n}|}$$

11. Basic Relationship 3: $\frac{1}{\ddot{a}_{\overline{n}|i}} = \frac{1}{\ddot{s}_{\overline{n}|i}} + d$



Consider a loan of 1, to be paid back over n years with equal annual payments P .

$$P \cdot \ddot{a}_{\overline{n}|i} = 1 \Rightarrow P = \frac{1}{\ddot{a}_{\overline{n}|i}}$$

$$D \cdot \ddot{s}_{\overline{n}|i} = 1 \Rightarrow D = \frac{1}{\ddot{s}_{\overline{n}|i}}$$

• Basic Relationship 4: Due = immediate $\times (1+i)$

$$\ddot{a}_{\overline{n}|} = \frac{1-v^n}{d} = \frac{1-v^n}{i} (1+i) = \underline{\underline{\ddot{a}_{\overline{n}|} \cdot (1+i)}}$$

$$\underline{\underline{\ddot{s}_{\overline{n}|}} = \frac{(1+i)^n - 1}{d} = \left[\frac{(1+i)^n - 1}{i} \right] (1+i) = \underline{\underline{s_{\overline{n}|} (1+i)}}$$

• Basic Relationship 5: $\ddot{a}_{\overline{n}|} = 1 + a_{\overline{n-1}|}$

$$\ddot{a}_{\overline{n}|} = 1 + (v + v^2 + \dots + v^{n-2} + v^{n-1})$$

$$= 1 + v \cdot (1 + v + \dots + v^{n-2})$$

$$= 1 + v \cdot \left(\frac{1-v^{n-1}}{1-v} \right) = 1 + \left(\frac{1}{1+i} \right) \left(\frac{1-v^{n-1}}{d} \right)$$

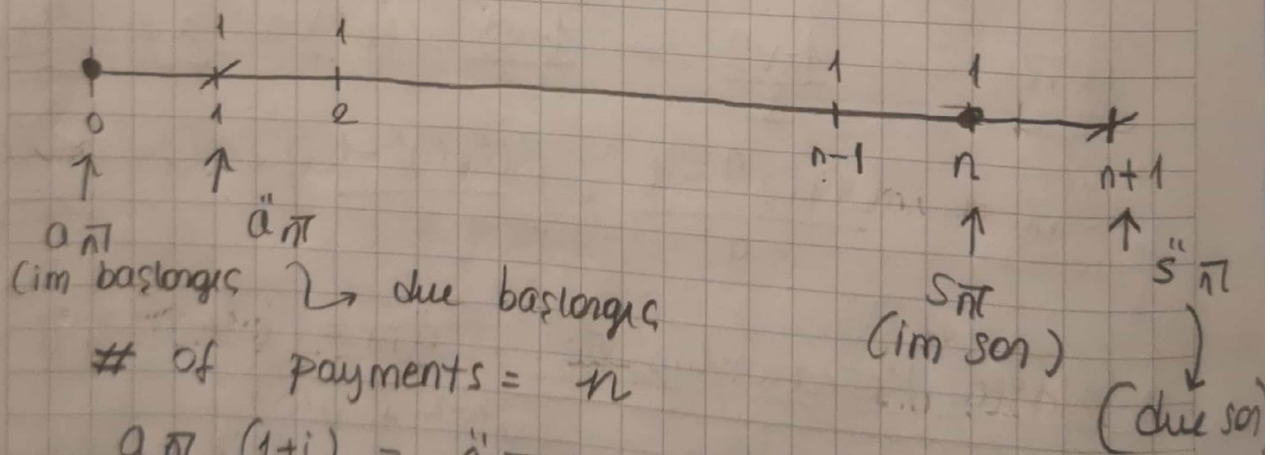
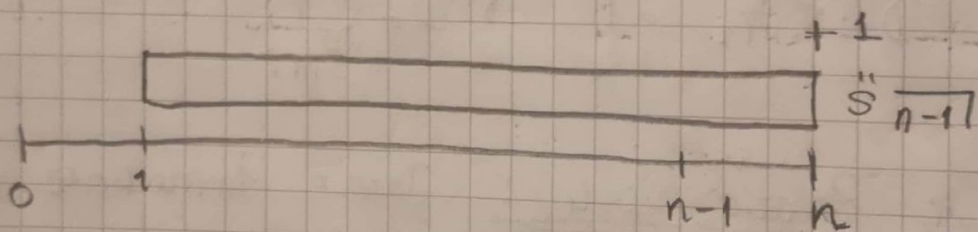
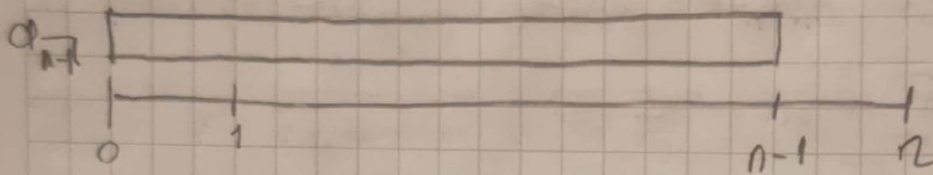
$$= 1 + \frac{1-v^{n-1}}{i} = 1 + a_{\overline{n-1}|}$$

• Basic Relationship 6: $S_{\overline{n}|i} = \ddot{S}_{\overline{n-1}|i}$

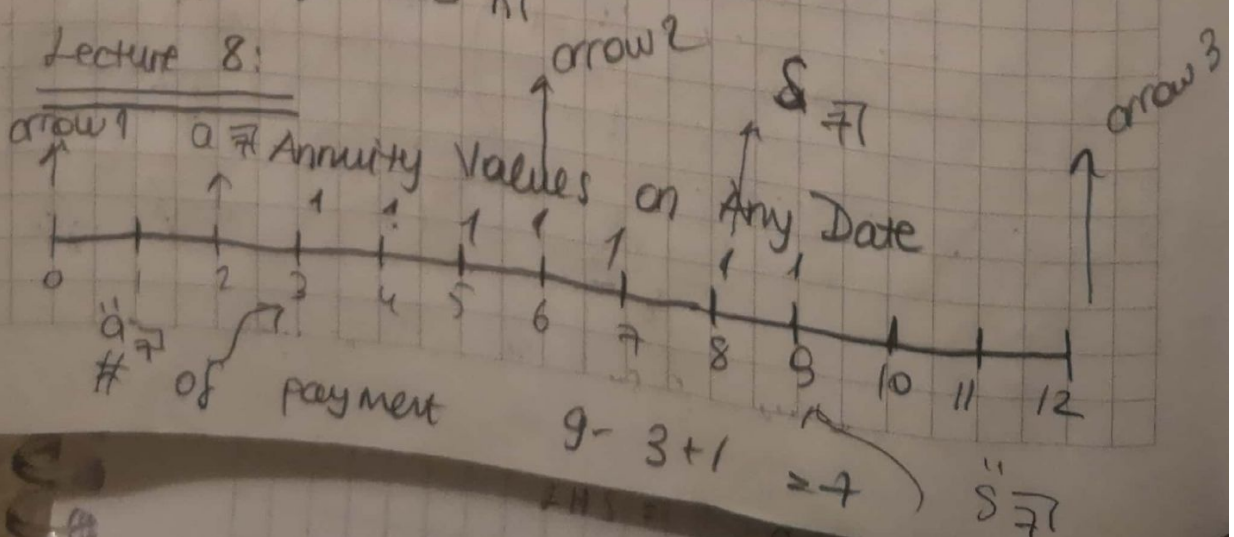
$$S_{\overline{n}|i} = [1 + (1+i) + (1+i)^2 + \dots + (1+i)^{n-1}]$$

$$= 1 + (1+i) \cdot [1 + (1+i) + \dots + (1+i)^{n-2}]$$

$$= 1 + (1+i) \cdot \left[\frac{(1+i)^{n-1} - 1}{i} \right] = 1 + \frac{(1+i)^{n-1} - 1}{d} = 1 + \ddot{S}_{\overline{n-1}|i}$$



Lecture 8:



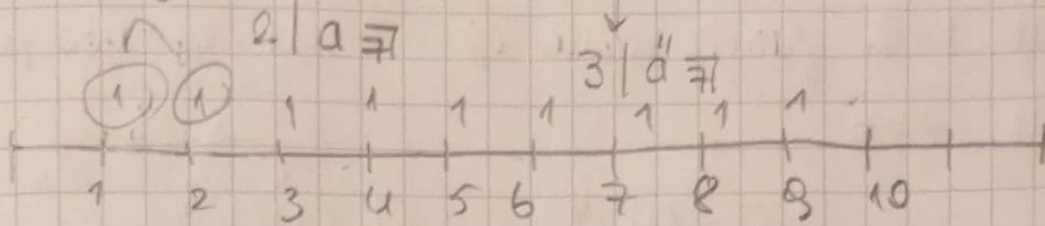
deferred for m periods
↑

It's called deferred annuity $\rightarrow m | a_{\overline{n}|}$

1st option

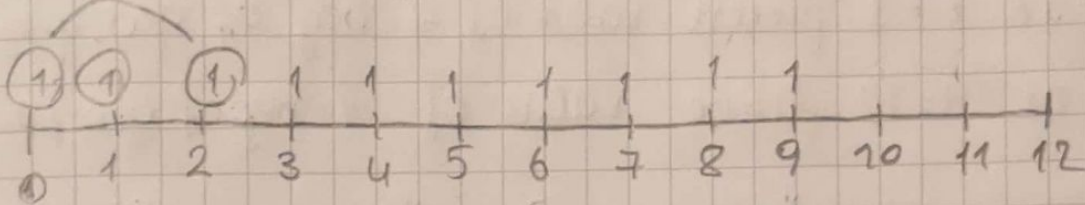
At time 0

hayali $a_{\overline{n}|} v^2 = a_{\overline{n}|} v^3$



$$a_{\overline{n}|} - a_{\overline{1}|} = v^2 \cdot a_{\overline{n}|}$$

hayali $v^m \cdot a_{\overline{n}|} = a_{\overline{m+n}|} - a_{\overline{m}|}$



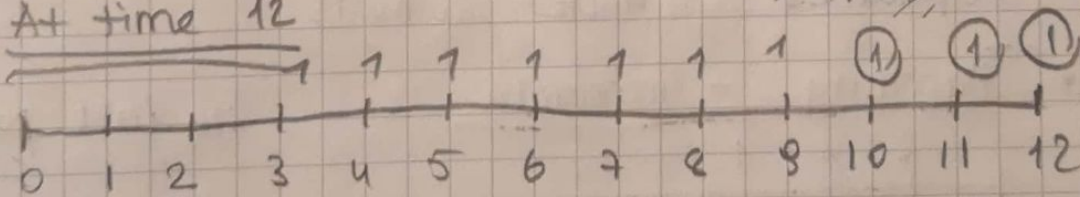
$$a_{\overline{10}|} - a_{\overline{3}|}$$

$$v^3 \cdot a_{\overline{n}|} = a_{\overline{10}|} - a_{\overline{3}|}$$

$$v^m \cdot a_{\overline{n}|} = a_{\overline{m+n}|} - a_{\overline{m}|}$$

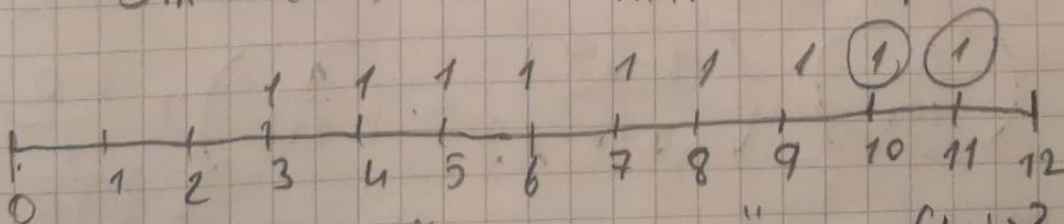
2nd option

At time 12



Ann im $S_{\overline{n}|} (1+i)^3 = S_{\overline{10}|} - S_{\overline{3}|}$

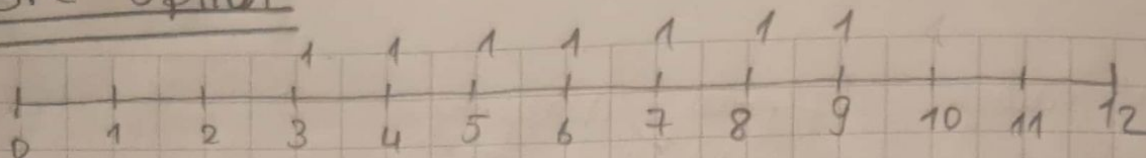
$$S_{\overline{n}|} \cdot (1+i)^m = S_{\overline{m+n}|} - S_{\overline{m}|}$$



$$S_{\overline{9}|} - S_{\overline{2}|} = S_{\overline{n}|} \cdot (1+i)^2$$

$$S_{\overline{n}|} \cdot (1+i)^m = S_{\overline{m+n}|} - S_{\overline{m}|}$$

3rd Option



⊛ at $t=2$ present value of 7 end-of-year pay = $a_{\overline{7}|i}$

at $t=9$ future value of 7 end-of-year pay = $s_{\overline{7}|i}$

at $t=6$

$$a_{\overline{7}|i} \cdot (1+i)^4 = s_{\overline{7}|i} (1+i)^{-3} \quad \xrightarrow{v^3}$$

$$a_{\overline{n}|i} \cdot (1+i)^m = v^{n-m} s_{\overline{n}|i}$$

⊛ at $t=3$ present value of 7 beg of year pay = $\ddot{a}_{\overline{7}|i}$

at $t=10$ future value of 7 beg of year pay = $\ddot{s}_{\overline{7}|i}$

$$\ddot{a}_{\overline{7}|i} (1+i)^3 = \ddot{s}_{\overline{7}|i} (1+i)^{-4} \quad \xrightarrow{v^4}$$

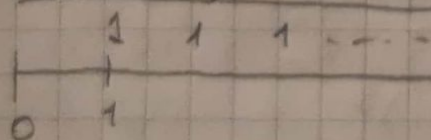
$$\ddot{a}_{\overline{n}|i} \cdot (1+i)^m = v^{(n-m)} \ddot{s}_{\overline{n}|i}$$

⊛

$$s_{\overline{n}|i} + a_{\overline{n}|i} = \ddot{s}_{\overline{n}|i} + \ddot{a}_{\overline{n}|i}$$

• Perpetuities: Infinite payments

• Perpetuity - Immediate:



no last payment

$$\textcircled{*} a_{\overline{\infty}|i} = 1(v) + (1)v^2 + (1)v^3 + \dots = v(1+v+\dots)$$

$$\left(\frac{1}{1+i} \right) \left(\frac{1-v^{\infty}}{1-v} \right) = \left(\frac{1}{1+i} \right) \times \frac{1-0}{\left(\frac{i}{1+i} \right)} = \frac{1}{i}$$

⊛ $s_{\overline{\infty}|i}$ is not defined.

$\frac{1}{i}$ represents an initial amount that can be invested at time = 0. if int is i

$$d'x = \frac{1}{d}$$

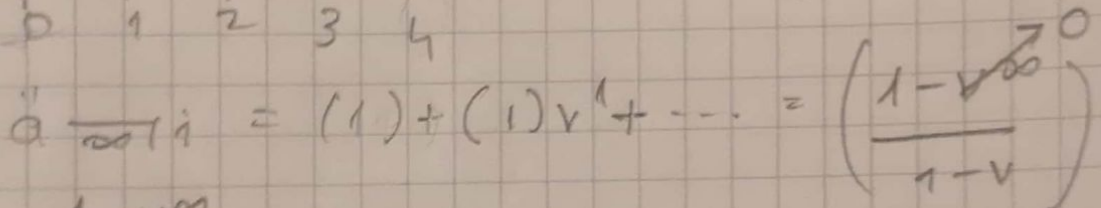
$$\frac{1}{i} \cdot i = 1$$

- $$a_n = a_{\infty} - v^n \cdot a_{\infty}$$

$$= a_{\infty} - v^n \cdot a_{\infty}$$

* Start after n years

- $$h \approx \infty$$



$$\frac{1 - v^\infty}{d} = \frac{1}{d}$$

~~$f, d \in I$~~

- $$\ddot{a} \overline{n} = \ddot{a} \overline{\infty} - v^n \cdot \ddot{a} \overline{1}$$

